A comprehensive study on fully convolutional networks in segmenting multi-anatomical structure of US medical images

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ABSTRACT

Ultrasound (US) imaging is widely used in clinical practice due to its safety, low cost, and real-time capability. However, accurate localization of anatomical structures remains challenging because of speckle noise, low contrast, and poorly defined boundaries. These challenges are amplified when segmenting multiple anatomical structures within a single scan. In this work, we present a comprehensive comparative study of fully convolutional and hybrid deep learning architectures for segmenting multi-anatomical structures in US images. As a part of this study, we curate a multi-anatomical US dataset by integrating the Open Kidney US and Spleenex datasets, resulting in 984 B-mode images with harmonized pixel-level annotations for five tissue types: capsule, cortex, medulla, central echo complex, and spleen. The dataset is processed using a uniform pipeline comprising consistent preprocessing, annotation harmonization, and structured data augmentation to maintain methodological consistency. Nine representative segmentation architectures—UNet, UNet++, Att-UNet, ResUNet++, Residual UNet, CMUNeXt, UCTransNet, MISSFormer, and EMCAD—are assessed under identical training protocols with a Dice–Focal composite loss function. Quantitative and qualitative results show that attention-based and multi-scale architectures consistently outperform conventional CNNs, with CMUNeXt achieving the highest Dice score of 0.7791 while maintaining a lightweight design. Performance degradation on small anatomical structures remains a key challenge, highlighting directions for future research in robust US segmentation.

1. Introduction

Ultrasound (US) radiology remains a highly clinician-dependent imaging modality, relying predominantly on the radiologist's visual interpretation and diagnostic expertise to accurately recognize organ structures, identify lesions and tissue types, assess bone structures, and perform precise anatomical measurements Klibanov and Hossack (2015). This manual approach offers the advantage of leveraging clinical experience and contextual understanding, allowing radiologists to adapt their assessments based on patient-specific variations MedScience (2025). However, indistinct tissue margins and the presence of speckle noise in US images make manual contouring difficult, often resulting in considerable intra- and inter-observer variation, thereby underscoring the necessity for automated segmentation Jiang, Guo, Bi, Huang, Yu and Wang (2023). Automatic tissue awareness in US imaging can significantly assist radiologists by providing consistent and objective identification of anatomical structures, thereby reducing observer variability and enhancing segmentation reproducibility Yang, Chen, Liang, Bai, Wang, Zhao, Ma et al. (2023). Ultrasound-specific artifacts, including speckle noise and boundary ambiguity, require advanced segmentation algorithms that can effectively model both fine-grained local features and broader contextual information.Xiao, Zhang, Shao, Liu, Shi, He and Kong (2025), Duque, Marquardt, Velikova, Lacourpaille, Nordez, Crouzier, Lee, Mateus and Navab (2024).

Multi-organ segmentation in US imaging has major therapeutic benefits since it allows for automated localization of organs such as the kidney, liver, and spleen, which aids in illness diagnosis, treatment planning, and longitudinal monitoring while reducing operator reliance. It also allows for quantitative assessment of organ size, texture, and pathology, which improves diagnostic accuracy and workflow efficiency. However, research in this subject still confronts significant shortcomings. US images contain speckle noise and poor contrast, making border recognition difficult; Organ shape variability and overlapping anatomical structures introduce additional complexity to the segmentation task. Existing models, which are frequently trained on restricted and single-center datasets, struggle to generalize across patient groups, diseases, and imaging technologies. Furthermore, there is a scarcity of large-scale, consistent, and well-annotated multi-organ US datasets, making technique benchmarking and comparison difficult.

While convolutional neural networks (CNNs), including U-Net and its variants, have demonstrated strong performance in segmenting individual organs, their effectiveness in multi-organ ultrasound segmentation is limited due to organ shape variability, overlapping structures, and heterogeneous tissue appearance. These challenges highlight the need for architectures capable of capturing both local details and contextual relationships across multiple anatomical regions. Litjens, Kooi, Bejnordi, Setio, Ciompi, Ghafoorian,

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Laak, Ginneken and Sánchez (2017),Chen, Li, Dai, Zhang and Yap (2022a).

Interpreting a full-body ultrasound scan is a highly challenging task, requiring simultaneous recognition of multiple organs, tissue types, and anatomical structures, which can vary widely across patients. Manual assessments are time-consuming, subjective, and prone to inter- and intra-observer variability, even for experienced radiologists. Recent advances in deep learning have shown promise in automating organ and tissue segmentation by capturing both global anatomical context and fine-grained tissue details. Motivated by these developments, our study focuses on systematically comparing representative deep learning architectures for multi-organ ultrasound segmentation. By evaluating these models on a unified multi-organ dataset, we aim to identify their relative strengths and limitations and provide guidance for future research in robust and clinically relevant multi-organ segmentation Wang, Wu and Liu (2025), Chen, Cai, Wang, Chen, Zhang, Han, Guo, Ding and Zhang (2024a), Liu, Qu, Xie, Shi and Song (2024).

This paper addresses the following research questions:
RQ1) How do fully convolutional architectures compare with hybrid deep learning architectures in segmenting multi-anatomical structures in US images?
RQ2) What are the relative strengths and weaknesses of existing deep learning models in handling complex anatomical structures in multi-anatomical US segmentation?
RQ3) what insights can be derived to guide future improvements in multi-anatomical ultrasound segmentation?
RQ4) How applicable are these deep learning architectures in real-world clinical settings?

The objectives of this study are: (i) *To conduct a comprehensive comparative analysis* of representative fully convolutional and hybrid deep learning architectures for multi-anatomical structure segmentation in ultrasound (US) images. (ii) *To curate and unify a standardized ultrasound dataset* by integrating the Open Kidney US and Spleenex datasets, resulting in 984 B-mode images with harmonized pixel-level annotations for five tissue types to ensure fair and reproducible benchmarking. (iii) *To apply consistent preprocessing, label unification, and data augmentation techniques* in order to establish a standardized experimental setting for objective model comparison. (iv) *To evaluate nine representative segmentation models* (UNet, UNet++, Att-UNet, ResUNet++, Residual UNet, CMUNeXt, UCTransNet, MISSFormer, and EMCAD) under identical training configurations using a combined loss function. (v) *To analyze the strengths, limitations, and practical implications* of these architectures and to provide clear future research directions for advancing multi-anatomical ultrasound segmentation.

The main contributions of this study are summarized as follows: (1) *Comprehensive Comparative Study:* A systematic analysis of fully convolutional and hybrid deep learning architectures for multi-anatomical ultrasound (US) segmentation under standardized training and evaluation settings. (2) *Clinically Motivated Curated Dataset:* A unified multi-anatomical US dataset of 984 B-mode images (534 kidney, 450 spleen) with harmonized annotations for five tissue types, enabling reproducible benchmarking and reflecting real clinical scenarios. (3) *Standardized Benchmarking:* Nine representative models are evaluated with identical training protocols using a combined loss function, providing fair quantitative and qualitative comparisons. (4) *Identification of High-Performing Architectures:* Attention-based and multi-scale models outperform conventional CNNs, with CMUNeXt achieving the highest Dice score of 0.7791 while maintaining a lightweight design. (5) *Highlighting Research Challenges and Future Directions:* Performance on small anatomical structures remains limited, indicating the need for noise-aware, multi-scale, and attention-based architectures, and larger, diverse multi-organ US datasets for future research.

This study offers a thorough evaluation of deep learning models for multi-organ US image segmentation, providing valuable insights into their performance, limitations, and future research directions, particularly emphasizing the need for improved handling of small structures and robust noise filtering.

The remainder of this paper is organized as follows. Section 2 reviews related work on ultrasound segmentation and attention/efficiency-aware architectures. Section 3 presents the materials and methods, including study design, dataset description and preprocessing, training and evaluation protocol, and comparative baseline. Section 4 details the experimental results, both quantitative and qualitative. Section 5 discusses the findings, conclusion, and future directions.

2. Related Works

Recently, deep learning has significantly improved the field of US medical image segmentation. Several studies have explored some innovative approaches to improve the US medical image segmentation.

2.1. U-Net Variants

The MSCA-UNet model was proposed by Chen et al. (2024) Chen, Zhu, Liu and Gao (2024b) for the segmentation of medical ultrasound images. In order to modify the channel weights using the channel attention method, the model has a multi-scale module that creates feature maps at different scales. The information at the edges is preserved using the global feature module. On several datasets, this network enhanced the IoU and dice score. These writers are aware of the issues with the loud photographs from the US. It is quite difficult to remove the borders of the lesions because of their lack of contrast and inconsistent grayscale. Additionally, there was a very tiny dataset of US pictures used.

The Attention and Unet models are specifically made for kidney segmentation MRI data with good boundary delineation. This architecture's features are extremely pertinent to the US medical picture segmentation challenge. However, it is limited in its ability to capture the organ boundary's low contrast and thick edge. Additionally, in US imaging, segmentation accuracy varies according to organ size Amiri, Brooks, Behboodi and Rivaz (2020).

A study Arora, Kumar and Soni (2024) on kidney segmentation in MRI, relevant due to similar boundary delineation challenges, evaluated U-Net and Attention U-Net with standard and focal cross-entropy losses, achieving a Dice coefficient of 0.966. Its emphasis on precise border segmentation applies to multi-organ US imaging, where overlapping organ structures create challenges.

In cardiovascular US imaging, an Attention-UNet model for carotid plaque segmentation achieved a correlation coefficient of 0.96 and an AUC of 0.97, surpassing U-Net, UNet++, and UNet3P models. Its ability to segment bright and fuzzy plaques highlights its potential for multi-organ US imaging, where variable tissue textures complicate segmentation Jain, Dubey, Saba, Khanna, Laird, Nicolaides and Sharma (2022).

Debesh Jha et al. Jha, Smedsrud, Riegler, Johansen, De Lange, Halvorsen and Johansen (2019) suggest ResUNet++ is an improved version of ResUNet designed for accurate polyp segmentation in colonoscopic images. It outperforms U-Net and ResUNet, achieving high Dice and mIoU scores on Kvasir-SEG and CVC-612 datasets. Its strong performance in handling complex tissue boundaries makes it promising for multi-organ US segmentation.

2.2. Hybrid Models (Convolutional and Transformer-based)

EMCAD Rahman, Munir and Marculescu (2024) architecture introduces an efficient attention decoder with various attention mechanisms for reducing the model complexity to segment the multi-organ on ultrasound images. It shows the segmentation accuracy of 83.63%, which can perform better than other state-of-the-art methods for multi-organ medical US image segmentation. But this study does not show performance on real-world clinical validation. This research also does not explain how the decoder can overcome the common clinical data challenges.

The CMUNeXt Tang, Ding, Quan, Wang, Ning and Zhou (2024) model is proposed to extract the global context by using the inductive bias. This architecture leverages a large kernel and an inverted bottleneck to mix distant spatial and local information thoroughly, which extracts the global context information efficiently. The CMUNeXt demonstrates that the outperforms state-of-the-art heavyweight and lightweight networks of medical images. This architecture is utilized by the small size of medical data.

The study Chen, Yin, Dai, Zhang, Yin and Cui (2022b) fully convolutional neural network (FCNN), originally for CT scans, was successfully adapted to T2-weighted MRI for multi-organ segmentation, showing high Dice scores and outperforming traditional methods. Its strong performance without modality-specific retraining highlights its robustness and potential for multi-organ US segmentation despite imaging challenges.

Another study Wu, Liu, Sun, Liu, Hou, Jiang and Zhang (2024) proposed a novel multi-scale, deep-supervised CNN for kidney segmentation in US images achieved high accuracy (Dice: 95.86%) by combining multi-scale inputs, pyramid pooling, and refined prediction through deep supervision. Its strong performance and robustness to imaging challenges make it well-suited for multi-organ US segmentation. BACANet Huang, Deng, Li and Yuan (2021) is a lightweight deep learning model designed for real-time liver US segmentation, combining efficiency with high accuracy. It uses convolutional attention, large kernel modules, and boundary supervision to enhance feature extraction and edge detection.

Tianyu Zhao et al. Wang, Zhang and Liu (2022) proposed A U-Net-based method for breast tumor US image segmentation integrates residual blocks and a fusion attention mechanism to enhance accuracy and stability. The residual blocks prevent gradient vanishing, while the attention mechanism improves feature extraction by focusing on spatial and channel information.

The MISSFormer Zhao and Dai (2022) and ConvTransNet Zhou, Hou, Lu, Zhao, Dang and Dong (2022) methods leverages the transformer-based architecture with CNN technologies to capture the global and local contexts effectively. These architectures potentially handled complex anatomical structures in US imaging. These architectures can not address the problem of feature discrimination limitation.

3. Materials and Methods

3.1. Study Design

This study is designed as a systematic comparative evaluation of deep learning architectures for multi-anatomical structure US segmentation. Nine representative models—UNet, UNet++, Att-UNet, ResUNet++, Residual UNet, CMUNeXt, UCTransNet, MISSFormer, and EMCAD—were selected to benchmark their performance under consistent experimental conditions. To ensure fair comparison, all models were trained and evaluated using identical protocols, including the same dataset splits, preprocessing steps, and optimization strategy.

A multi-anatomical US dataset was curated by integrating the Open Kidney Ultrasound (OKUS) and Spleenex datasets, comprising a total of 984 B-mode ultrasound images annotated for five tissue types: capsule, cortex, medulla, central echo complex, and spleen. The dataset was standardized through intensity normalization and harmonized annotations to enable reproducible benchmarking. Data augmentation techniques, including random rotation, horizontal and vertical flips, and intensity scaling, were applied during training to increase robustness against ultrasound-specific variations and improve generalization.

All models were trained using a Dice–Focal composite loss function to address class imbalance and optimize segmentation accuracy. The Adam optimizer was used with an initial learning rate of 0.001. Training was performed for 400 epochs with a batch size of 8, and early stopping was applied based on validation loss to prevent overfitting.

All experiments were conducted on a high-performance workstation equipped with an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory, ensuring efficient training

and reproducibility. This study design provides a controlled framework to systematically evaluate model performance, identify strengths and limitations, and guide future research in multi-organ ultrasound segmentation.

3.2. Dataset Description and Preparation

The Open Kidney Ultrasound (OKUS) Singla, Ringstrom, Hu, Lessoway, Reid, Nguan and Rohling (2023) Dataset, a comprehensive, high-quality collection of kidney B-mode ultrasound images. Compiled over five years, the dataset includes scans from over 500 adult patients with common conditions such as diabetes mellitus, immunoglobulin A nephropathy, and hypertension. It encompasses both native and transplanted kidneys and features detailed multi-class semantic segmentation annotations generated by two expert sonographers. The cohort has an average age of 53.2 ± 14.7 years, with 63% male and 37% female participants. The majority of images (91.2%) are of fair or good quality. The Spleenex Dataset Karimi, Seraj, Mirzadeh Sarcheshmeh, Fazli, Seraj, Eslami, Khanmohamadi, Sajjadian Moosavi, Ghattan Kashani, Sajjadian Moosavi and Shariat Panahi (2025) comprises 450 anonymized US images aimed at advancing spleen segmentation for medical diagnostics, particularly for conditions like splenomegaly. Collected using Philips Affinity 70 and Supersonic Aixplorer devices, the images are converted from DICOM to 512×512 PNG format. with intensity normalized to a [0, 1] range for consistency. Ground truth segmentation Masks were meticulously crafted using 3DSlicer software through a collaborative effort. involving two radiologists and four computer engineering students, with radiologists' verification ensuring accuracy. The dataset strictly adheres to ethical standards, with patient anonymity preserved and informed consent obtained

To support comprehensive segmentation of kidney and spleen tissue types, the OKUS (534 images) and Spleenex Dataset (450 images) were merged, resulting in a collection of 984 B-mode US images. All images are stored in 512 × 512 PNG format with intensities normalized to the [0, 1] range. The resulting Combined Multi-Anatomy US Dataset forms a robust foundation for developing and evaluating deep learning models for kidney and spleen segmentation, although additional validation is advised to address device-specific variations and ensure clinical generalizability.

All images were resized to 256 × 256 pixels and normalized to the range $[0, 1]$. The final dataset contains K anatomical classes, including background, and all ground-truth labels were converted to one-hot encoded masks to support multi-class learning consistency. Image intensities were normalized using dataset-specific mean and standard deviation values to ensure consistent intensity ranges and stable training. To enhance model robustness and mitigate overfitting, synchronized data augmentation was applied to both images and masks, including random rotations up to ±15°, horizontal flips, and vertical flips with a 50% probability each, while preserving perfect spatial correspondence between anatomical structures and their labels. This standardized preprocessing pipeline ensures dataset uniformity, increases anatomical diversity, and establishes a reproducible foundation for comparative evaluation of deep learning models in multi-anatomical structure ultrasound segmentation.

Figure 1 illustrates the distribution of the multi-anatomical US dataset. The outer ring shows the contribution of the OKUS and Spleenex datasets, while the inner ring represents tissue types: Capsule (green), Central Echo Complex (blue), Medulla (yellow), Cortex (red), and Spleen (magenta). Percentages indicate relative proportions.

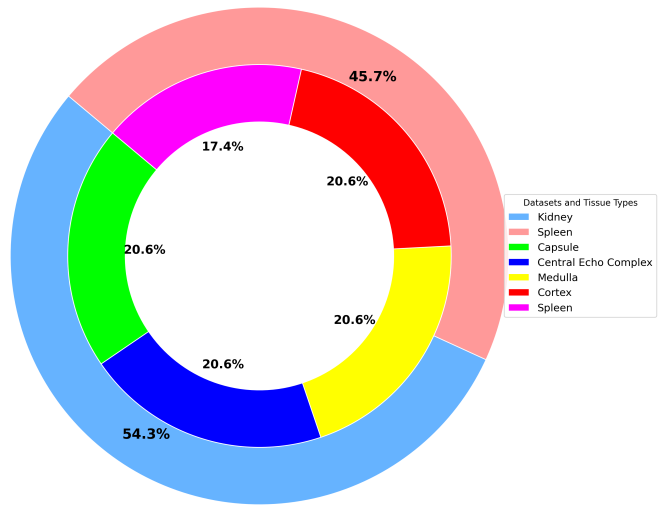

Figure 1: Dataset and tissue type distribution in the Multi-Anatomical US Dataset.

3.3. Training and Evaluation Protocol

The training of the model was carried out using the Adam optimizer with an initial learning rate of 0.001. A batch size of 8 was used, and training was conducted for 400 epochs. We also employed early stopping based on the validation loss to mitigate overfitting. To ensure unbiased evaluation, the dataset was split into training (80%), testing (10%) and validation (10%) sets at the image level, with no patient-level overlap between splits. Table 1 represents the hyperparameter settings used for the comparative evaluation.

3.3.1. Loss Function

To address class imbalance and boundary precision in multi-class segmentation, the training objective combines Dice loss and focal loss. Let $\hat{Y}_{i,j,k}$ and $Y_{i,j,k}$ represent the predicted probability and ground-truth label for class k at pixel (i, j), in the corresponding order.

The multi-class soft Dice loss is defined as

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{K} \sum_{k=1}^{K} \frac{2 \sum_{i,j} \hat{Y}_{i,j,k} Y_{i,j,k} + \epsilon}{\sum_{i,j} \hat{Y}_{i,j,k} + \sum_{i,j} Y_{i,j,k} + \epsilon}, \quad (1)$$

where ϵ is a small constant for numerical stability.

The focal loss is given by

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{HW} \sum_{i,j} \sum_{k=1}^{K} Y_{i,j,k}(1-\hat{Y}_{i,j,k})^{\gamma} \log(\hat{Y}_{i,j,k}), \quad (2)$$

where γ controls the focusing strength.

The final training loss is computed as

$$\mathcal{L} = \alpha\mathcal{L}_{\text{Dice}} + (1-\alpha)\mathcal{L}_{\text{Focal}}, \quad (3)$$

with $\alpha = 0.7$ in all experiments.

3.3.2. Comparative Baseline

To evaluate the performance of deep learning architectures on multi-anatomical structure ultrasound segmentation, nine representative models were selected as comparative baselines. These models span both fully convolutional and hybrid designs, incorporating attention mechanisms and multi-scale feature representations.

- **UNet:** A classical encoder-decoder architecture with skip connections that preserves spatial information for precise segmentation. It serves as a widely used baseline for biomedical image segmentation.
- **UNet++:** An extension of UNet that introduces nested and dense skip connections to improve feature propagation and model accuracy, particularly for fine-grained structures.
- **Att-UNet:** Integrates attention gates within the UNet framework to focus on relevant regions, enhancing the segmentation of small or complex anatomical structures.
- **ResUNet++:** Combines residual connections with UNet++'s nested skip connections, improving gradient flow and learning efficiency for deeper networks.
- **Residual U-Net:** Incorporates residual blocks into the UNet architecture to facilitate training of deeper networks and improve feature learning.
- **CMUNeXt:** A lightweight, multi-scale convolutional architecture designed for efficient segmentation, balancing high accuracy with low computational cost.
- **UCTransNet:** A hybrid convolution-transformer architecture that leverages both local convolutional features and global context modeling via transformer blocks.
- **MISSFormer:** A transformer-based architecture emphasizing multi-scale self-attention, capable of capturing long-range dependencies and anatomical context.
- **EMCAD:** An ensemble-based multi-scale convolutional approach integrating channel and spatial attention, designed for robust segmentation of heterogeneous structures.

Table 1: Training and hyperparameter settings used for the comparative evaluation of deep learning models in multi-anatomical structure US segmentation.

Hyperparameter	**Value**
Optimizer	Adam
Learning rate	0.001
Learning scheduler	Not used / default (Adam step-size control)
Loss function	Composite loss: Dice + Focal ($\alpha = 0.7$, $\gamma = 2.0$)
Number of epochs	400
Batch size	8
Input dimension	$256 \times 256 \times 3$ (RGB)
Hidden dimension	ResNet-34 backbone (64, 128, 256, 512 filters)
Number of classes	5 (Capsule, Central Echo Complex, Medulla, Cortex, Spleen)
Random seed	2222

These models were selected to provide a comprehensive benchmark, covering classical CNNs, attention-enhanced networks, and transformer-based architectures, enabling systematic analysis of their strengths, limitations, and suitability for multi-anatomical structure segmentation in US imaging.

4. Results

This section presents the quantitative and qualitative performance of the selected deep learning models for multi-anatomical structure segmentation in ultrasound (US) images, following the experimental protocol described in the Materials and Methods. All results are reported on the validation split using identical preprocessing, training configurations, and the Dice–Focal composite loss function, ensuring a fair and consistent comparison across all models.

4.1. Evaluation Metrics

The Segmentation effectiveness was calculated using standard pixel-wise and region-based measures to thoroughly assess classification accuracy and anatomical segmentation standard. Below is the list of utilized metrices list:

- **Dice Similarity Coefficient (Dice):** A region-based overlap metric measuring segmentation quality for each class. Higher values indicate better agreement between predicted and ground truth masks.
- **Loss:** The Dice–Focal composite loss used during training to optimize segmentation performance and handle class imbalance.

- **Number of Parameters (M):** The total number of trainable parameters (in millions) for each model, reflecting model complexity and computational cost.
- **Intersection over Union (IoU):** A region-based metric quantifying the overlap between predicted and ground truth masks; higher values indicate more accurate segmentation.

All presented metrics are computed on the test set and aggregated across classes unless otherwise specified.

4.2. Overall Quantitative Results

Table 2 presents a comparative evaluation of different segmentation models on the combined multi-anatomy ultrasound dataset using Dice Score, Loss, Intersection-over-Union (IoU), and model parameter count as performance indicators. Higher Dice Score and IoU, along with lower Loss, denote superior segmentation quality, while the number of parameters reflects computational complexity. Among all evaluated methods, CMUNeXt achieves the best overall performance with the highest Dice Score of 0.7791, the highest IoU of 0.6323, and the lowest Loss of 0.0814, while maintaining a lightweight architecture with only 3.14 M parameters, demonstrating an excellent accuracy–efficiency trade-off. UCTransNet and ResUNet++ also deliver strong segmentation performance with Dice Scores of 0.7648 and 0.7617, respectively; however, UCTransNet incurs significantly higher computational cost due to its large parameter count (59.74 M). Standard convolutional architectures such as Att-UNet, UNet, and UNet++ achieve competitive Dice Scores in the range of 0.7517–0.7540, but with comparatively higher Loss values, indicating minor inconsistencies in boundary delineation. In contrast, MISSFormer, Residual U-Net, and EMCAD exhibit noticeably lower Dice Scores (0.7028, 0.6412, and 0.6170, respectively) and higher Loss values, suggesting reduced robustness in handling ultrasound-specific artifacts such as speckle noise and low contrast. Overall, the results demonstrate that attention-based and hybrid architectures, particularly those combining multi-scale feature extraction with efficient design, provide superior segmentation accuracy and reliability for multi-organ ultrasound imaging, supporting their potential for clinical deployment.

4.3. Per-Tissue Dice Score Analysis

Table 3 presents a per-tissue Dice score comparison of different segmentation models across five anatomical structures in ultrasound imaging, namely Capsule, Central Echo Complex, Medulla, Cortex, and Spleen, along with the average Dice score across all classes. Among all evaluated methods, CMUNeXt achieves the highest average Dice score (0.7791), demonstrating consistently strong performance across all tissue types, particularly excelling in the segmentation of the Central Echo Complex and Spleen, with Dice scores of 0.8359 and 0.9834, respectively. Attention-based and hybrid architectures, including UCTransNet, ResUNet++, Att-UNet, UNet++, and UNet, also exhibit robust segmentation capability, with average Dice scores ranging from 0.7517 to 0.7648, indicating reliable multi-tissue delineation. These models show notably high accuracy in well-defined regions such as the Central Echo Complex (Dice > 0.82) and Spleen (Dice > 0.93), while performance in the Cortex and Medulla remains comparatively lower, reflecting the difficulty of accurately segmenting subtle boundaries and low-contrast regions inherent to ultrasound data. Transformer-based models such as MISSFormer achieve moderate performance with an average Dice score of 0.7028, but show reduced accuracy in more complex tissue regions, suggesting limitations in capturing fine-grained spatial details. In contrast, Residual U-Net and EMCAD yield lower average Dice scores (0.6412 and 0.6170, respectively), indicating challenges in maintaining consistent anatomical delineation across tissue types. Overall, the results confirm that architectures integrating attention mechanisms and multi-scale convolutional features are more effective in handling the heterogeneous and noise-prone characteristics of ultrasound images, leading to improved and more stable multi-anatomical segmentation performance.

4.4. Qualitative Results

Figure 2 illustrates the qualitative segmentation results on the kidney US dataset using baseline and attention-enhanced convolutional architectures. Attention-based models, such as Att-UNet and CMUNeXt, demonstrate strong visual agreement with the ground truth by accurately delineating kidney anatomical boundaries and producing coherent, well-localized tissue regions with minimal leakage. In comparison, conventional architectures such as UNet and UNet++ exhibit slightly irregular boundaries and mild under-segmentation in regions with complex textures and low contrast, indicating limitations in preserving fine structural details in challenging kidney US images.

Figure 3 further evaluates the kidney US dataset, focusing on advanced and hybrid architectures, including UCTransNet, ResUNet++, MISSFormer, Residual UNet, and EMCAD. Transformer-based and hybrid models show improved spatial consistency, enhanced contextual modeling, and refined boundary delineation compared to traditional CNN-based methods. Although occasional boundary distortions and minor misclassifications are observed in highly heterogeneous regions, these models generally achieve superior structural continuity and spatial accuracy, highlighting the benefits of attention mechanisms and advanced feature fusion strategies for kidney tissue-type segmentation.

Figure 4 presents a qualitative visual comparison of tissue-type segmentation results on the spleen US dataset using additional fully convolutional network (FCN)-based models. While these architectures are capable of capturing the overall spleen anatomy, they often struggle with precise boundary delineation due to speckle noise, weak contrast, and heterogeneous tissue appearance inherent in spleen US images. Fragmented predictions, boundary smoothing, and region leakage are particularly noticeable along organ boundaries. Compared to the attention-enhanced and

Table 2
Segmentation performance comparison of different methods on the combined multi-anatomy ultrasound dataset.

Methods	Dice Score	Loss	Parameters (M)	IoU
CMUNeXt	**0.7791**	**0.0814**	3.14	**0.6323**
UCTransNet	0.7648	0.0836	59.74	0.6203
ResUNet++	0.7617	0.0970	4.06	0.6138
UNet++	0.7540	0.1590	9.16	0.6178
Att-UNet	0.7518	0.0879	34.87	0.6066
UNet	0.7517	0.1393	13.39	0.6219
MISSFormer	0.7028	0.1105	30.05	0.5717
Residual U-Net	0.6412	0.1940	13.04	0.5497
EMCAD	0.6170	0.1420	**2.57**	0.4975

Table 3
Per-Tissue Dice Score Comparison of Different Segmentation Methods for Multi-Anatomical Structures in US Imaging.

Methods	Capsule	Central Echo Complex	Medulla	Cortex	Spleen	Average
CMUNeXt	0.7207	0.8359	0.6832	0.6724	0.9834	0.7791
UCTransNet	0.7129	0.8433	0.6463	0.6413	0.9801	0.7648
ResUNet++	0.7054	0.8299	0.6661	0.6301	0.9771	0.7617
UNet++	0.7123	0.8197	0.6427	0.6244	0.9709	0.7540
Att-UNet	0.7208	0.8390	0.6399	0.6248	0.9346	0.7518
UNet	0.6946	0.8566	0.6454	0.6296	0.9325	0.7517
MISSFormer	0.6522	0.7653	0.5773	0.5936	0.9258	0.7028
Residual U-Net	0.5517	0.7605	0.6028	0.5892	0.7016	0.6412
EMCAD	0.5711	0.7125	0.5376	0.4907	0.7729	0.6170

transformer-based models evaluated on the kidney dataset (Figs. 11 and 12), standard FCN-based approaches demonstrate reduced sensitivity to fine structural details and global contextual information.

Overall, the qualitative comparisons across Figures 2–4 consistently demonstrate that models incorporating attention mechanisms, transformer components, and advanced feature fusion strategies provide more accurate and robust tissue-type segmentation in US imaging. These visual observations strongly complement and support the quantitative results reported in the study for both kidney and spleen datasets.

5. Discussion

As illustrated in Figure 5, t-SNE visualizations are employed to qualitatively analyze and compare the discriminative power of deep feature representations learned by different segmentation models. The t-SNE technique projects high-dimensional feature spaces into two dimensions while preserving local neighborhood structures, thereby enabling intuitive assessment of inter-class separability and intra-class compactness.

The feature embeddings generated by Att-UNet exhibit relatively compact and well-separated clusters, indicating strong discriminative capability among different tissue types. This behavior reflects the effectiveness of the attention mechanism in highlighting anatomically relevant regions and suppressing irrelevant background features, which is particularly beneficial for ultrasound images characterized by speckle noise and low contrast. Similarly, CMUNeXt and UCTransNet demonstrate improved clustering behavior compared to conventional CNN-based models, suggesting enhanced feature fusion and global context modeling. Their embeddings show clearer separation between tissue classes, although moderate overlap remains in regions corresponding to anatomically ambiguous boundaries. In contrast, the baseline UNet and Residual U-Net exhibit more scattered distributions with noticeable inter-class overlap, implying limited feature discrimination. This overlap is indicative of challenges in capturing fine-grained anatomical details and complex texture variations inherent in ultrasound data using standard convolutional operations alone. Advanced architectures such as UNet++, ResUNet++, MISSFormer, and EMCAD show varying degrees of improvement. Transformer-based and hybrid models (e.g., MISSFormer and EMCAD) generally yield better spatial grouping and improved class separability, benefiting from long-range dependency modeling. However, some cluster dispersion persists, suggesting

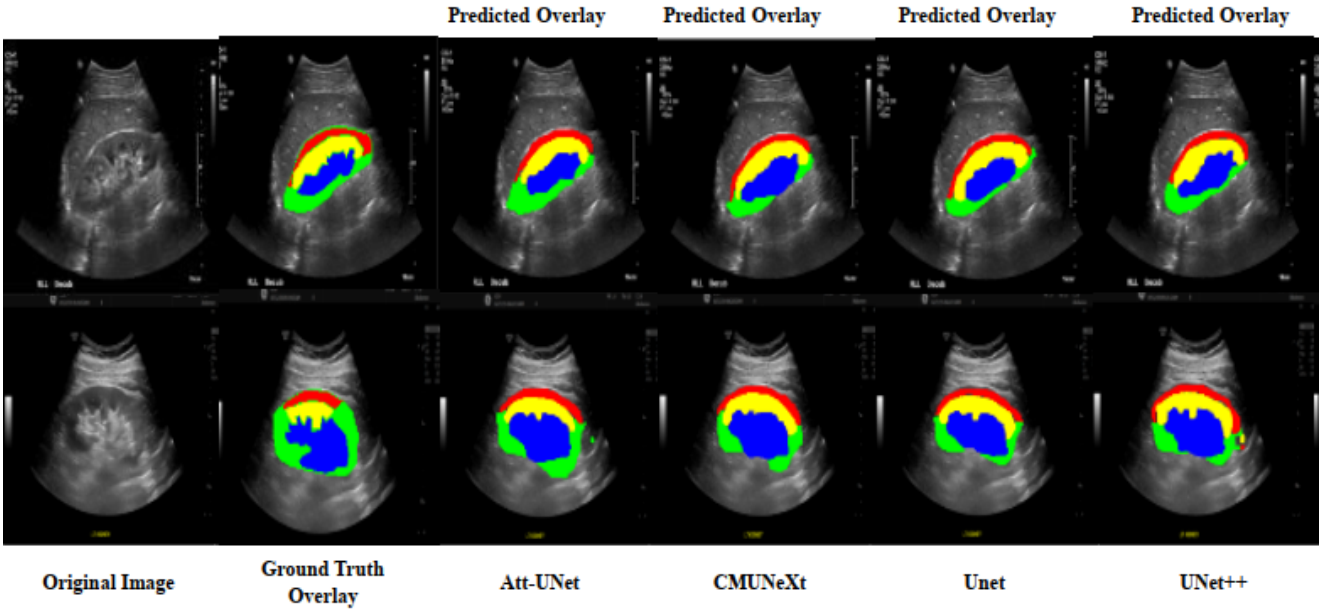

Figure 2: Visual comparison of tissue-type segmentation on kidney scans using Att-UNet, CMUNeXt, UNet, and UNet++. Capsule (green), Central Echo Complex (blue), Medulla (yellow), and Cortex (red) are shown to highlight segmentation differences across models, especially in small and challenging anatomical regions.

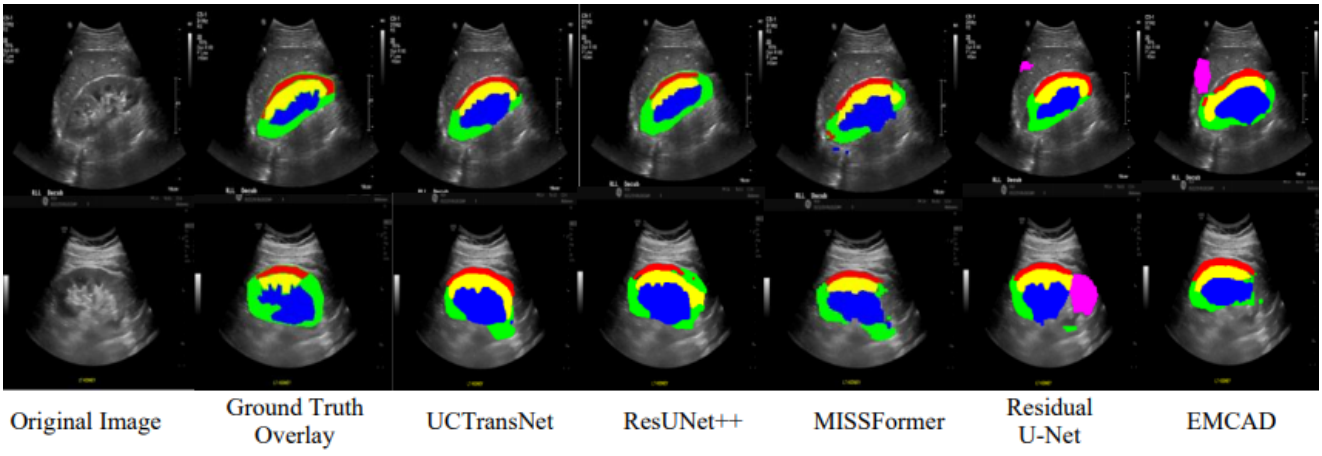

Figure 3: Visual comparison of tissue-type segmentation on kidney scans using UCTransNet, ResUNet++, MISSFormer, Residual UNet, EMCAD. Capsule (green), Central Echo Complex (blue), Medulla (yellow), and Cortex (red) are shown to highlight segmentation differences across models, especially in small and challenging anatomical regions.

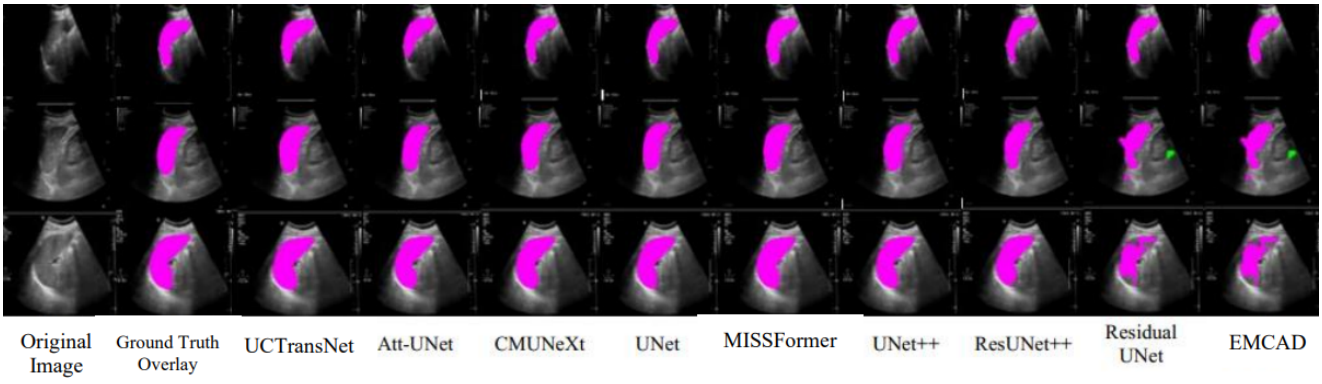

Figure 4: Visual comparison of tissue-type segmentation on spleen scans using fully convolutional networks. Spleen is shown in magenta color to highlight segmentation performance, emphasizing the strengths and limitations of each architecture.

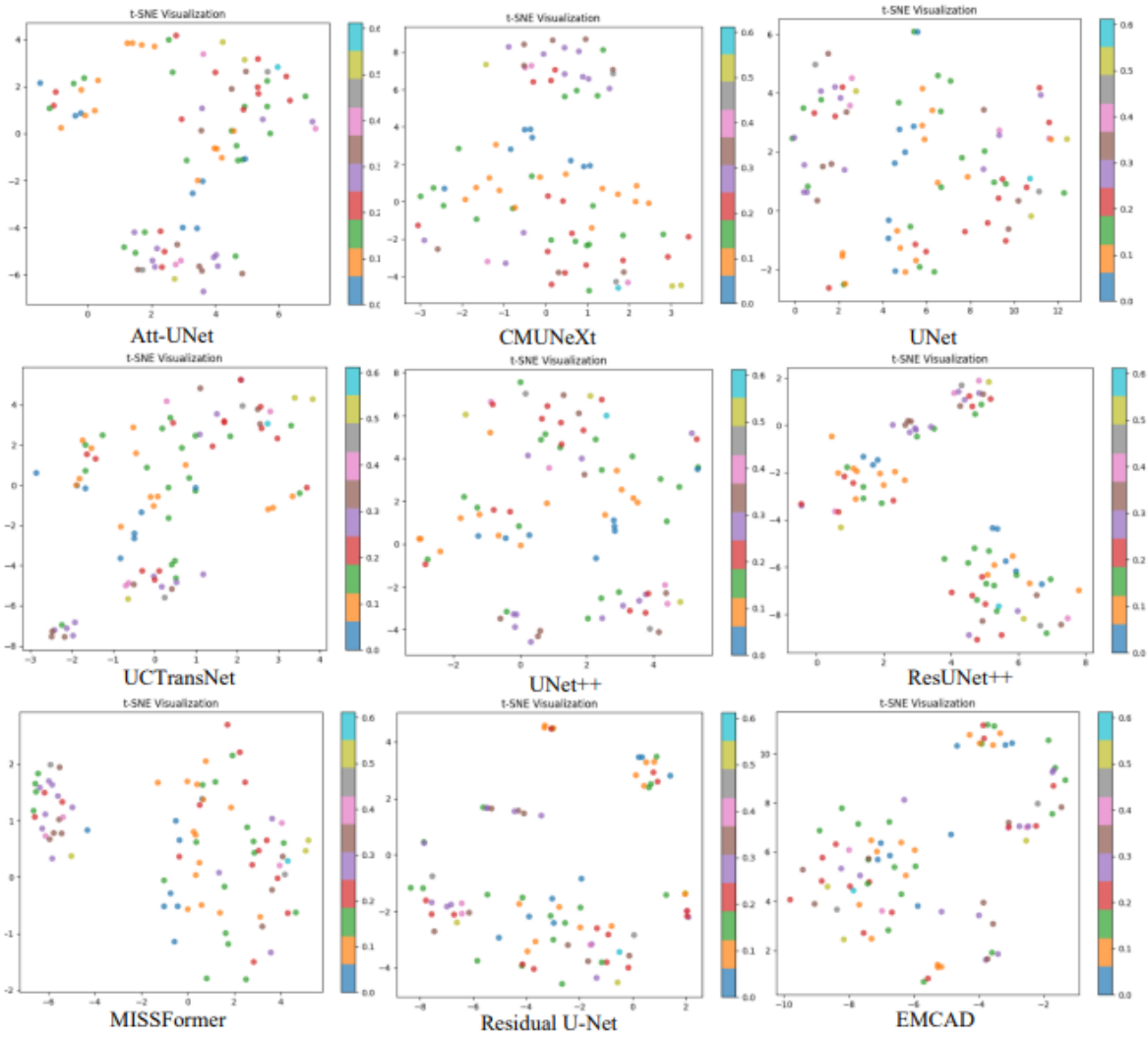

Figure 5: t-SNE visualization of deep feature embeddings learned by different segmentation models.

sensitivity to heterogeneous tissue appearances and boundary uncertainty.

Overall, the t-SNE analysis provides strong qualitative evidence that models incorporating attention mechanisms, multi-scale feature aggregation, and advanced contextual modeling learn more robust and semantically meaningful feature embeddings. These visual observations are consistent with the quantitative segmentation results reported earlier and further confirm the superiority of attention- and transformer-enhanced architectures for multi-anatomical US image segmentation.

Figure 6 provides a detailed area-wise analysis of segmentation performance by reporting the average Dice score across different ground-truth size bins for all evaluated models. For extremely small anatomical regions (0–50, 51–100, and 101–250 pixels), all methods achieve Dice scores close to zero, indicating that reliable segmentation of very small tissue structures remains highly challenging in US images due to severe speckle noise, low signal-to-noise ratio, and poorly defined boundaries. Once the region size exceeds 250 pixels, a pronounced performance gain is observed across all architectures, reflecting improved spatial coherence and more stable feature representation. In the medium-size range (251–500 and 501–1k pixels), Dice scores rise to approximately 0.45–0.60, with attention-based models such as Att-UNet, CMUNeXt, and ResUNet++ consistently outperforming conventional U-Net variants by better suppressing background clutter and emphasizing anatomically relevant features. For large anatomical structures (1k–5k and 5k+ pixels), performance further improves and saturates, with Dice scores exceeding 0.65 for most models, demonstrating that sufficient spatial context enables accurate shape and boundary modeling. Transformer-based and hybrid models show competitive performance for larger regions but remain less effective for smaller structures, likely due to their reliance on global context rather than fine-grained local cues. Overall, this figure highlights object size as a critical factor in multi-anatomical US segmentation and reveals that, despite strong performance on medium-to-large

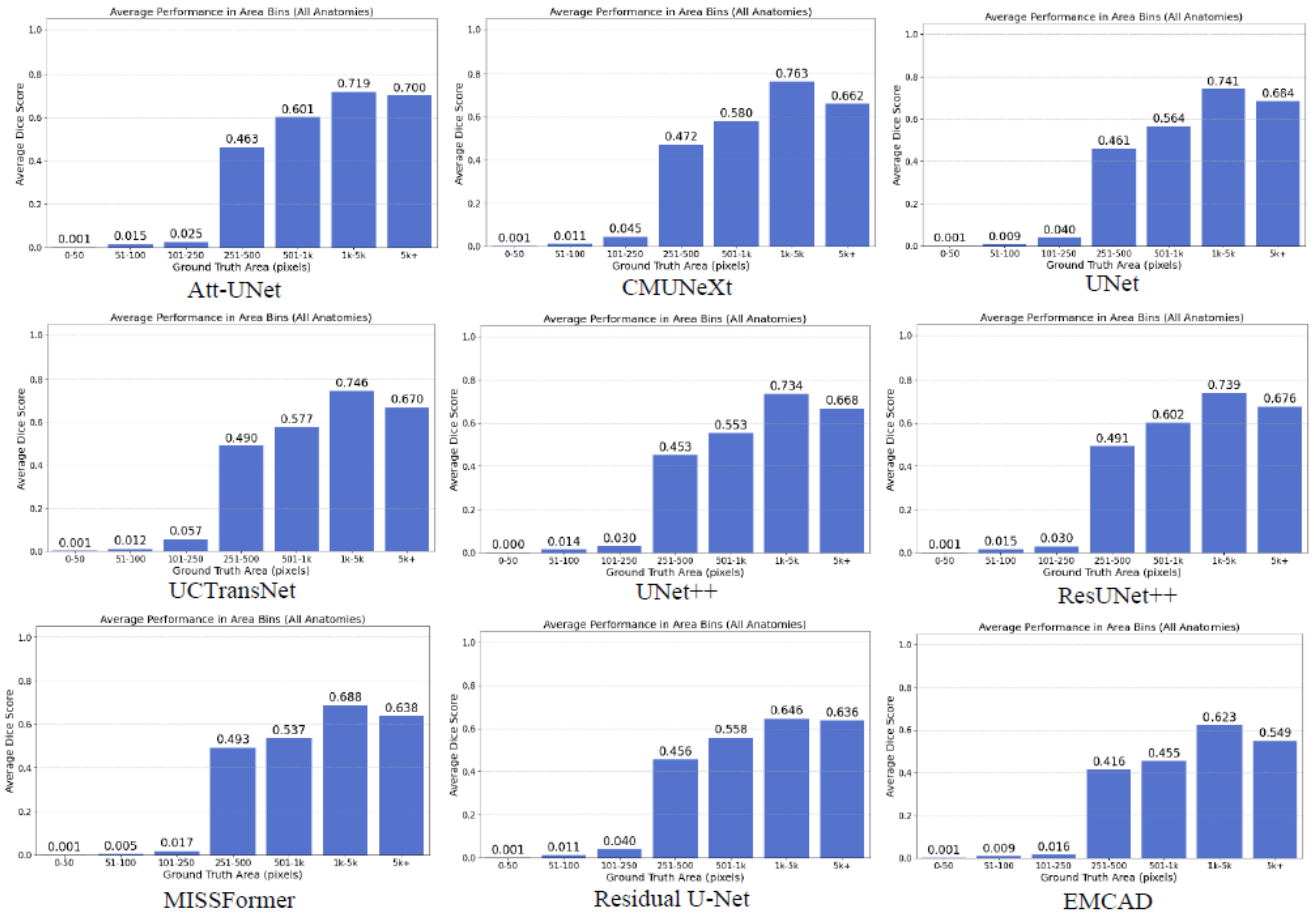

Figure 6: Area-wise segmentation performance of different Fully Convolutional Networks on the multi-anatomical ultrasound dataset.

structures, accurate delineation of small anatomical regions remains an open research challenge.

6. Conclusion and Future Work

In this study, we conducted a comprehensive evaluation of state-of-the-art fully convolutional and hybrid deep learning architectures for multi-anatomical structure segmentation in ultrasound (US) imaging. By integrating the Open Kidney Ultrasound (OKUS) and Spleenex datasets, we curated a multi-anatomical US dataset consisting of 984 B-mode images with harmonized pixel-level annotations for five anatomical classes: capsule, cortex, medulla, central echo complex, and spleen. This unified dataset, combined with a standardized preprocessing pipeline and consistent training protocol, enabled fair, reproducible, and unbiased benchmarking across nine representative segmentation models.

The experimental results provide several important insights. First, architectures incorporating attention mechanisms and multi-scale feature aggregation consistently outperformed conventional convolutional models in both quantitative and qualitative evaluations. In particular, CMUNeXt achieved the highest overall Dice score (0.7791) and IoU (0.6323) while maintaining a lightweight architecture (3.14M parameters), demonstrating an excellent balance between segmentation accuracy and computational efficiency. This makes it particularly suitable for real-time or resource-constrained clinical environments, such as portable ultrasound systems.

Second, transformer-based and hybrid convolution–transformer architectures showed improved contextual modeling and feature discrimination compared to traditional CNNs. However, their performance gains were sometimes offset by increased computational complexity and parameter counts, which may limit deployment in low-resource clinical settings.

Third, per-tissue and area-wise analyses revealed that segmentation accuracy strongly depends on anatomical structure size. While medium-to-large regions achieved satisfactory Dice scores, all models struggled with extremely small anatomical structures (0–250 pixels), where Dice scores approached zero. This performance degradation highlights the persistent challenges posed by ultrasound-specific artifacts such as speckle noise, low contrast, acoustic shadowing, and ambiguous tissue boundaries. Even advanced attention-based models remain sensitive to these imaging limitations.

The t-SNE feature visualization further confirmed that attention-enhanced and multi-scale models learn more discriminative and semantically meaningful feature embeddings, resulting in improved inter-class separability. Nevertheless, overlapping clusters in complex regions indicate that anatomical variability and boundary ambiguity remain unresolved issues.

Future work will focus on two key directions. First, we plan to construct a large-scale and diverse multi-organ ultrasound dataset encompassing additional abdominal and soft-tissue organs, varied patient demographics, and scans acquired from multiple vendors and clinical centers. Such diversity is essential to improve model generalization and clinical reliability. Second, future research will emphasize the development of more reliable and robust segmentation architectures tailored to US imaging, capable of effectively handling speckle noise, anatomical variability, and inter-organ appearance differences. This includes exploring advanced attention mechanisms, noise-aware feature learning, and hybrid convolution–transformer designs. Together, these efforts aim to move toward clinically deployable ultrasound segmentation systems capable of expert-level multi-organ interpretation.

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